Complete EEG DSP System Notes (STM32F446RE + CMSIS-DSP + ESP12E + BIOAMP-EXG pill)

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04/12/2025

Section 0 — Project Purpose

This system is a fully functional embedded EEG processing engine built on the STM32F446RE (ARM Cortex-M4F). Its goals:

- Deterministic EEG sampling at **512 Hz**.
- Preprocessing using **50 Hz notch** and **0.5–70 Hz bandpass FIR**.
- Splitting signal into **8 clinical EEG subbands**.
- Extraction of **25+ EEG features**: time, frequency, fractal, autoregressive, and cepstral.
- Hybrid rule-based seizure detection.
- UART JSON telemetry with **CRC16-MODBUS** reliability.

Architecture mirrors medical EEG DSP pipelines:

Acquisition → Preprocessing → Feature Extraction → Detection → Telemetry

Section 0.1 — BioAmp EXG Pill for EEG Acquisition

The BioAmp EXG Pill is the analog front-end (AFE) used to interface biological signals (EEG, ECG, EMG) with the STM32 ADC. It converts microvolt-level scalp potentials into a clean, conditioned, amplified signal that a 12-bit ADC can safely digitize.

Although extremely small, the BioAmp EXG Pill implements the complete chain of biopotential amplification used in medical instrumentation.

0.1.1 Purpose of the AFE

Raw EEG at the scalp is extremely weak:

$$\text{Amplitude} = 10\text{–}100\ \mu\text{V}, \qquad \text{Frequency} = 0.5\text{–}70\ \text{Hz}$$

Without amplification and filtering:

- the ADC would read **flat zero** most of the time,
- noise would drown out the EEG,
- DC offsets would saturate the ADC rails,
- mains interference (50 Hz) would dominate the signal.

Thus the BioAmp EXG Pill performs:

Amplification + Filtering + Level Shifting

before sending the conditioned signal to the microcontroller.

0.1.2 Internal Architecture

The BioAmp EXG Pill implements a compact instrumentation acquisition chain:

1. **Instrumentation Amplifier Stage** High CMRR (> 100 dB) rejects common-mode noise (50 Hz hum, motion artifact, electrode drift). Gain is typically:
$$G_1 \approx 21\text{–}100$$
2. **Active Filtering Stage** Provides:
   - high-pass filtering at ≈ 0.5 Hz to remove electrode DC drift,
   - low-pass filtering at ≈ 80 Hz to suppress EMG and RF noise.
3. **Precision Level Shifting** EEG can be negative, but the STM32 ADC can only measure:
$$0\ \text{V} \le V_{\text{ADC}} \le 3.3\ \text{V}$$
The amplifier shifts the signal up to mid-supply:
$$V_{\text{center}} = 1.65\ \text{V}$$
so that the EEG oscillates around the ADC midpoint.
4. **Output Scaling / Gain Compensation** The final gain is tuned so that typical EEG (∼ 100 μV) becomes:
$$100\ \mu\text{V} \times G_{\text{total}} \approx 0.3\text{–}1.0\ \text{V}_{\text{pp}}$$
which fits well inside the ADC dynamic range.

This produces a clean signal that the digital pipeline can interpret accurately.

0.1.3 Why the BioAmp EXG Pill Works Well for EEG

Several design choices make it especially effective for EEG:

(a) High Input Impedance Scalp electrodes have impedance in the range:

$$5\text{–}50\ \text{k}\Omega$$

The EXG Pill input impedance (> 10 MΩ) ensures that:

- electrode loading is negligible,
- signal distortion is minimized,
- noise pickup is reduced.

(b) Excellent CMRR The instrumentation amplifier rejects common-mode noise:

$$\text{CMRR} > 100 \text{ dB}$$

This is critical because:

- EEG amplitudes are microvolt-level,
- mains interference is millivolt-level.

(c) Low Noise Design The AFE uses:

- low-noise op-amps,
- optimized gain staging,
- carefully chosen capacitors/resistors,

making it suitable for microvolt signals.

(d) Integrated Biasing and Reference Driving The 1.65 V reference is clean, stable, and low noise. Without this:

- the ADC would saturate,
- the digital pipeline would see large DC offsets,
- filters would behave incorrectly.

0.1.4 Output Characteristics

The EXG Pill outputs a unipolar signal:

$$0.3 \text{ V} \approx V_{\text{min}} \leq V_{\text{OUT}} \leq V_{\text{max}} \approx 3.0 \text{ V}$$

This range matches perfectly with:

$$\text{STM32 ADC input range: 0–3.3 V}$$

After ADC conversion:

$$v_{\text{ADC}} = \frac{4095}{3.3} \cdot V_{\text{out}}$$

Digital post-processing subtracts the 1.65 V offset (Section 2).

0.1.5 Advantages Compared to DIY Analog Front-Ends

- **Microsized**, fits directly onto wearable EEG form factors.
- **Highly stable** amplification chain.
- **Low-noise layout**, avoiding PCB mistakes.
- **Ready to use**, no need for analog circuit debugging.
- Provides **reliable amplitude scaling** for ADC consumption.

For embedded EEG systems, this dramatically shortens development time while preserving high signal fidelity.

0.1.6 Summary of BioAmp EXG Pill Role in the System

BioAmp EXG Pill converts microvolt EEG → millivolt-level ADC-r

It ensures:

- amplification of weak scalp EEG,
- noise rejection through high CMRR,
- filtering of drift and high-frequency noise,
- safe level shifting for ADC,
- stable, low-noise EEG capture.

This makes the downstream digital pipeline (TIM2, ADC1, DMA, CMSIS DSP, FFT, seizure detection, etc.) possible and reliable.

Section 1 — EEG Acquisition (TIM2 + ADC1 + DMA)

This section explains **how the microcontroller collects EEG data** in a precise and continuous manner. Even if you are new to embedded systems, the goal here is to make the idea clear.

1.1 Deterministic Sampling (Why TIM2 is needed)

In EEG processing, the **timing of every sample matters**. If the sampling rate wobbles or drifts, the frequency information becomes distorted, and the FFT results become unreliable.

To prevent this, we use the STM32 hardware timer TIM2 to generate a sampling tick at a fixed rate:

$$f_s = 512 \text{ Hz}$$

This means:

$$\text{TIM2 produces one trigger every } \frac{1}{512} \text{ seconds}$$

Each trigger tells the ADC:

"Take one sample *now*, at this exact moment."

Because the timer runs in hardware:

- The gaps between samples are **exactly equal**.
- There is **no randomness** (called "jitter").
- The sampling frequency never drifts.

Why does this matter?

- EEG is heavily analyzed in the frequency domain (FFT, bandpowers).
- For FFT to be meaningful, samples must be evenly spaced.

Thus, TIM2 ensures:

Clean and accurate frequency analysis.

1.2 Using ADC + DMA (Why DMA is needed)

When the ADC converts signals to digital numbers, the values must be stored somewhere. A naive beginner approach is:

> "Let the CPU read each ADC value one-by-one inside a loop."

This does **not** work for real-time EEG because:

- The CPU would be too slow.
- Interrupts could be missed.
- Data would be lost.

Therefore we use DMA (Direct Memory Access).

DMA allows the ADC to store incoming samples **directly into RAM** without bothering the CPU.

We configure DMA to store samples into a buffer of size:

$$1024 = 2 \times 512$$

This is called a **double buffer** or **ping-pong buffer**:

$$\underbrace{512 \text{ samples}}_{\text{first half}} \quad | \quad \underbrace{512 \text{ samples}}_{\text{second half}}$$

How the half-buffer system works

The DMA generates two interrupts:

$$\text{Half Complete} \Rightarrow \text{"first 512 samples are ready"}$$

$$\text{Complete} \Rightarrow \text{"second 512 samples are ready"}$$

This means:

- While DMA fills the **first half**, the CPU processes the **second half**.
- While DMA fills the **second half**, the CPU processes the **first half**.

This overlap is why it is called **ping-pong buffering**:

$$\text{DMA writes} \longleftrightarrow \text{CPU reads and processes}$$

They never block each other. No samples are lost.

Processing Deadline

You receive 512 samples every:

$$\frac{512}{512 \text{ Hz}} = 1 \text{ second}$$

This means:

The CPU has exactly 1 second to finish processing a block.

As long as the DSP pipeline finishes before the next block arrives, **real-time EEG processing is achieved**.

Section 2 — ADC Conversion and Scaling

In this section we explain how the raw EEG voltage (coming from the analog front-end) is converted into digital values by the ADC, and then mathematically scaled into meaningful units.

Even if you are a beginner, the ideas here are fundamental to any biomedical signal acquisition system.

2.1 How the ADC Represents Voltage

The STM32F446RE uses a **12-bit ADC**, meaning it can represent analog signals using:

$$2^{12} = 4096 \text{ discrete levels}$$

So the ADC output value v will always lie within:

$$v \in [0, 4095]$$

This number does NOT represent volts directly—it is only a digital code.

The ADC internally compares the input voltage to a reference voltage:

$$V_{\text{REF}} = 3.3 \text{ V}$$

Thus the smallest possible step (the resolution) is:

$$\Delta V = \frac{3.3}{4095} \approx 0.0008 \text{ V} = 0.8 \text{ mV}$$

2.2 Convert ADC Code to Actual Voltage

To convert the ADC reading v into voltage V:

$$V = \frac{v}{4095} \times 3.3$$

This gives the **true analog input voltage at the ADC pin**.

2.3 Why There is a Mid-Supply Offset (1.65 V)

Real EEG signals are extremely small:

$$\text{Amplitude: } 10\text{–}100\ \mu\text{V}$$

The analog front-end (instrumentation amplifier) shifts the EEG signal upwards so that it is centered at:

$$\frac{3.3}{2} = 1.65 \text{ V}$$

Why? Because the ADC cannot read negative voltages. All EEG voltages must be kept within the safe ADC range $[0, 3.3]$.

Therefore, the raw ADC voltage looks like:

$$V = 1.65 \text{ V} + (\text{EEG in volts})$$

2.4 Removing the Offset

To reconstruct the real bipolar EEG around zero, we subtract the mid-supply offset:

$$x[n] = V - 1.65$$

This yields:

- Negative values when EEG goes below baseline
- Positive values when EEG rises above baseline

Now $x[n]$ is a **zero-centered EEG signal in volts**, ready for filtering and DSP.

2.5 Why Centering is Essential

Many DSP operations assume signals are centered around zero:

- FFT assumes zero-mean (improves accuracy)
- FIR filters behave correctly with bipolar signals
- Entropy, variance, AR models use deviations from mean

If we did not subtract 1.65 V, then:

- Variance would be artificially huge
- RMS would be dominated by the DC offset
- FFT would contain a strong peak at 0 Hz

Thus, this step is crucial for all later EEG processing.

Section 3 — Preprocessing and Filtering

Before extracting features from EEG, the signal must be cleaned. Raw EEG contains:

- Mains noise (50 Hz in India)
- High-frequency interference
- Slow drift and DC offsets

This section explains the three major filters used in the system:

Notch filter → Wide bandpass FIR → Subband FIRs

All filtering is done in floating point using CMSIS-DSP on the Cortex-M4F.

3.1 50 Hz Notch Filter (IIR Biquad)

The purpose of this filter is to reduce mains interference at 50 Hz. Instead of removing a large range of frequencies, we "notch out" only a very narrow band around 50 Hz.

Design Parameters

$$f_0 = 50\,\text{Hz}, \quad Q = 30, \quad f_s = 512\,\text{Hz}$$

Normalized frequency

$$\omega_0 = \frac{2\pi f_0}{f_s}$$

Bandwidth term The quality factor Q sets how narrow the notch is:

$$\alpha = \frac{\sin(\omega_0)}{2Q}$$

Digital Biquad Transfer Function The filter is implemented as a second-order section:

$$H(z) = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{1 + a_1 z^{-1} + a_2 z^{-2}}$$

The CMSIS-DSP library processes this efficiently using the "Direct Form II Transposed" structure.

Why a Notch Filter?

- EEG signals are extremely weak (microvolts)
- A 50 Hz spike can dominate the entire spectrum
- Notch filtering removes the interference with minimal distortion of nearby EEG bands

This is standard in every medical EEG amplifier.

3.2 301-Tap Bandpass FIR (0.5–70 Hz)

After removing mains noise, we apply a very clean wide bandpass FIR filter. The purpose is to isolate the "legal" EEG band:

0.5 Hz to 70 Hz

This FIR has:

- Linear phase (no waveform distortion)
- Very steep cutoff
- Very low ripple in passband and stopband

Ideal Bandpass Impulse Response For an ideal rectangular bandpass filter, the impulse response is:

$$h_{\text{ideal}}[n] = 2f_2 \operatorname{sinc}(2f_2(n - M/2)) - 2f_1 \operatorname{sinc}(2f_1(n - M/2))$$

Where:

$$f_1 = \frac{0.5}{f_s}, \qquad f_2 = \frac{70}{f_s}$$

$$M = N - 1 \quad (\text{for } N = 301 \text{ taps})$$

Applying a Kaiser Window The ideal filter has infinite length; windowing forces it to finite length.

Kaiser window:

$$w[n] = \frac{I_0\left(\beta\sqrt{1-\left(\frac{2n}{N-1}-1\right)^2}\right)}{I_0(\beta)}$$

where:

- β controls side-lobe suppression
- I_0 is the modified Bessel function of the first kind

Final FIR Coefficients

$$h[n] = h_{\text{ideal}}[n] \cdot w[n]$$

These taps are fed into `arm_fir_f32()` for efficient convolution.

Why 301 Taps?

- Longer filter = sharper cutoff
- Controls artifacts before FFT
- Reproduces EEG spikes with perfect phase lineraity

3.3 Eight Subband FIR Filters

Each subband isolates a classical EEG rhythm band. These are all 129-tap FIR filters with linear phase.

Band	Frequency Range (Hz)
0	0.5–4 (Delta)
1	4–8 (Theta)
2	8–10 (Low Alpha)
3	10–13 (High Alpha)
4	13–20 (Low Beta)
5	20–30 (Mid Beta)
6	30–45 (Low Gamma)
7	45–70 (High Gamma)

Why We Use Subbands Each EEG rhythm corresponds to different brain states:

- Delta: deep sleep, some seizure morphologies
- Theta: drowsiness, focal seizure activity
- Alpha: occipital rhythm (eyes closed)
- Beta/Gamma: high activity, often increases during seizures

Subband filtering allows:

- Per-band energy computation
- Detecting specific frequency-domain seizure biomarkers
- Independent analysis of rhythms

Why 129 Taps?

- Enough taps for decent transition bandwidth
- Efficient on Cortex-M4F (129 is small compared to 301)
- Maintains linear phase for spike preservation

Summary of Section 3 By the end of preprocessing:

- 50 Hz noise is removed
- The EEG is restricted to the medically meaningful range
- Clean subband components are available

This prepares the signal for fast, stable feature extraction.

Section 4 — FFT and Spectral Processing

After filtering, the EEG signal is ready for frequency-domain analysis. The Cortex-M4F performs a **512-point Real FFT (RFFT512)** every second using CMSIS-DSP.

This converts the time-domain EEG into its frequency components, which allows us to compute:

- Power Spectral Density (PSD)
- Band powers (Delta, Theta, Alpha, Beta)
- Dominant frequency
- Spectral centroid
- Spectral flatness

These are essential biomarkers used in clinical EEG interpretation.

4.1 RFFT512 (Real Fast Fourier Transform)

The FFT size is:

$$N = 512$$

Sampling rate:

$$f_s = 512 \text{ Hz}$$

Frequency Resolution Each bin represents a 1 Hz band:

$$\Delta f = \frac{f_s}{N} = \frac{512}{512} = 1 \text{ Hz}$$

This means:

- Bin 1 = 1 Hz
- Bin 10 = 10 Hz
- Bin 30 = 30 Hz

Output of the Real FFT For a real-valued signal, the FFT output is packed as:

$$\text{DC},\ \text{Nyquist},\ \Re[1], \Im[1],\ \Re[2], \Im[2],\ \ldots$$

Power Spectral Density (PSD) To compute power at each frequency bin:

$$P[k] = \Re[k]^2 + \Im[k]^2$$

This gives the energy distribution across frequencies.

4.2 Band Powers (Delta, Theta, Alpha, Beta)

The PSD values are summed across clinically meaningful EEG bands.

$$\text{Delta (0.5–4 Hz)} = \sum_{k=f=0.5}^{4} P[k]$$

$$\text{Theta (4–8 Hz)} = \sum_{k=f=4}^{8} P[k]$$

$$\text{Alpha (8–13 Hz)} = \sum_{k=f=8}^{13} P[k]$$

$$\text{Beta (13–30 Hz)} = \sum_{k=f=13}^{30} P[k]$$

Why Band Power Matters Each band corresponds to a brain rhythm:

- Delta — deep sleep, slow-wave activity
- Theta — drowsiness, focal seizure patterns
- Alpha — visual cortex rhythm, suppressed in seizures
- Beta — high activity, increases during ictal bursts

Band power changes are one of the strongest indicators of abnormal EEG states.

4.3 Spectral Features

These quantify how the energy is distributed across the frequency spectrum.

Dominant Frequency The strongest (highest-power) frequency:

$$f_{\text{dom}} = \arg\max_k P[k]$$

This reflects the main oscillatory rhythm in the EEG block.

Spectral Centroid The "center of gravity" of the frequency spectrum:

$$f_c = \frac{\sum_k f[k]\, P[k]}{\sum_k P[k]}$$

High centroid values indicate higher-frequency rhythmic activity.

Spectral Flatness Measures "tone-like" vs "noise-like" structure:

$$\text{SF} = \frac{\exp\left(\frac{1}{N}\sum_k \ln P[k]\right)}{\frac{1}{N}\sum_k P[k]}$$

Interpretation:

- SF $\approx 1 \rightarrow$ signal is noise-like (e.g., EMG artifact)
- SF $\approx 0 \rightarrow$ signal is tonal/rhythmic (e.g., seizure activity)

Summary of Section 4 Using FFT and PSD:

- We find how much energy the EEG has in each frequency band.
- We detect strong rhythmic activity via dominant frequency.
- We compute advanced statistical descriptors (centroid, flatness).

These spectral metrics are essential inputs to the seizure detection algorithm.

Section 5 — Time-Domain EEG Features

Time-domain features describe the raw waveform itself, without transforming it into the frequency domain. These features capture **amplitude**, **shape**, **smoothness**, and **rhythmicity** of the EEG signal.

Let the signal block be:

$$x[n], \quad n = 0, 1, \ldots, N-1$$

Below are the major time-domain features extracted from every 1-second (512-sample) block.

5.1 Root Mean Square (RMS)

RMS measures the **effective amplitude** of the EEG. It increases strongly during seizures, where signal amplitude becomes high and rhythmic.

$$\text{RMS} = \sqrt{\frac{1}{N}\sum_{n=0}^{N-1} x[n]^2}$$

Interpretation:

- Low RMS $\rightarrow$ calm background EEG
- High RMS $\rightarrow$ bursting/spiking or ictal activity

5.2 Variance

Variance measures the **spread or energy** of the signal around its mean.

$$\sigma^2 = \frac{1}{N}\sum_{n=0}^{N-1}(x[n]-\mu)^2 \quad \text{where } \mu = \frac{1}{N}\sum x[n]$$

Characteristics:

- Highly elevated during generalized tonic–clonic seizures
- Indicates sudden energy bursts

5.3 Zero Crossings

Zero crossings count how often the signal changes sign:

$$x[n-1]\cdot x[n] < 0 \Rightarrow \text{one crossing}$$

Meaning:

- High zero-cross count → fast oscillations
- Low zero-cross count → slow waves (delta/theta)

During seizures with fast rhythmic activity, zero-crossing count increases.

5.4 Peak-to-Peak Amplitude

Measures the **range** of the signal in the block.

$$P2P = \max(x[n]) - \min(x[n])$$

Large P2P indicates:

- Sharp spikes
- Artifact bursts
- Ictal spike–wave complexes

5.5 Crest Factor

Crest factor describes how "spiky" the waveform is. It compares the maximum absolute deviation to the RMS value.

$$\text{Crest} = \frac{\max|x[n]|}{\text{RMS}}$$

Interpretation:

- Large crest factor → sharp spikes (common in epilepsy)
- Low crest factor → smooth sinusoidal waves

5.6 Higher-Order Statistical Moments

These measure **shape characteristics** of the EEG waveform.

Let:

$$m_k = \frac{1}{N}\sum_{n=0}^{N-1}(x[n]-\mu)^k$$

Skewness Skewness measures asymmetry of the waveform:

$$\text{Skew} = \frac{m_3}{m_2^{3/2}}$$

Meaning:

- Positive skew → more upward spikes
- Negative skew → more downward spikes
- Large |Skew| → non-Gaussian behavior (common in abnormal EEG)

Kurtosis Kurtosis measures how "peaked" or "heavy-tailed" the signal is:

$$\text{Kurt} = \frac{m_4}{m_2^2} - 3$$

Interpretation:

- Kurt $= 0$ → Gaussian-like signal
- Kurt > 0 → sharp spikes / epileptiform events
- Kurt < 0 → flatter, less peaky shapes

Summary of Time-Domain Features

- RMS and Variance quantify energy.
- Zero crossings quantify rhythmicity.
- Peak-to-peak and Crest describe spike sharpness.
- Skewness and Kurtosis capture waveform shape.

Together, these features offer a complete picture of the EEG's raw temporal behavior and provide crucial biomarkers for seizure detection.

Section 6 — Hjorth Parameters

Hjorth Parameters are classical EEG descriptors introduced in 1970. They summarise the **shape, smoothness,** and **complexity** of the EEG waveform in the time domain.

They are computed using the signal $x[n]$, its first derivative $\Delta x[n]$, and second derivative $\Delta^2 x[n]$.

Let:

$$\Delta x[n] = x[n] - x[n-1]$$
$$\Delta^2 x[n] = \Delta x[n] - \Delta x[n-1]$$

The three parameters are:

6.1 Activity

Activity measures the **signal variance**, which corresponds to its "power" or mean amplitude level.

$$\text{Activity} = \text{Var}(x) = \frac{1}{N}\sum_{n=0}^{N-1}(x[n]-\mu)^2$$

Interpretation:

- High Activity → high-amplitude EEG (common during seizures)
- Low Activity → calm resting-state EEG

6.2 Mobility

Mobility measures how **quickly the signal moves**, i.e., the relative amount of slope or "speed" present in the waveform. It is defined as the square root of the ratio of variances of the first derivative and the original signal.

$$\text{Mobility} = \sqrt{\frac{\text{Var}(\Delta x)}{\text{Var}(x)}}$$

Interpretation:

- High Mobility → rapid oscillations (theta/beta/gamma), seizure-like patterns
- Low Mobility → slow waves (delta)

6.3 Complexity

Complexity compares the Mobility of the first derivative to the Mobility of the signal itself.

It measures **how much the signal resembles a pure sine wave**. A pure sine wave has Complexity ≈ 1. More irregular or chaotic signals have higher Complexity.

$$\text{Complexity} = \frac{\text{Mobility}(\Delta x)}{\text{Mobility}(x)}$$

Expanded form:

$$\text{Complexity} = \frac{\sqrt{\frac{\text{Var}(\Delta^2 x)}{\text{Var}(\Delta x)}}}{\sqrt{\frac{\text{Var}(\Delta x)}{\text{Var}(x)}}}$$

Interpretation:

- Complexity ≈ 1 → smooth, sinusoidal-like EEG
- Complexity > 1 → irregular, chaotic EEG (early seizure onset often shows this)

Summary

- **Activity** measures amplitude power.
- **Mobility** measures the signal's "speed" or frequency content.
- **Complexity** measures deviation from a smooth sinusoid.

Together, Hjorth Parameters provide powerful descriptors of EEG dynamics and are widely used in seizure detection, BCI systems, and EEG-based classification tasks.

Section 7 — Fractal Features

Fractal features measure the **complexity**, **self-similarity**, and **irregularity** of EEG signals. Epileptic events often change the "fractal dimension" of the EEG, because seizures make the signal more rhythmic and less complex.

Two widely used fractal measures are:

- Higuchi Fractal Dimension (HFD)
- Petrosian Fractal Dimension (PFD)

Both are used in clinical EEG analysis and real-time seizure detection systems.

7.1 Higuchi Fractal Dimension (HFD)

Higuchi's method measures how the EEG curve "fills" the time axis at different scales. It computes the average curve length $L(k)$ for multiple scale factors k.

For each scale $k = 1, 2, \ldots, K_{\max}$:

$$L(k) \sim k^{-D}$$

Thus the fractal dimension D is obtained from the slope of the log–log plot:

$$D = -\text{slope}\,(\log L(k) \text{ vs. } \log k)$$

Interpretation of D:

- $D \approx 1.0$ → smooth, low-complexity signal (similar to a sine wave)
- D increases as the signal becomes more irregular
- During seizures, D often **drops** because EEG becomes more periodic and rhythmic

HFD is one of the strongest features for distinguishing normal vs. ictal EEG.

7.2 Petrosian Fractal Dimension (PFD)

Petrosian FD is a faster and simpler fractal measure based on zero-crossing behavior.

Let:

- N = number of samples
- Z = number of zero crossings (sign changes)

Petrosian FD is:

$$\text{PFD} = \frac{\log N}{\log N + \log\left(\frac{N}{N+0.4Z}\right)}$$

Interpretation:

- If the signal changes direction frequently (more zero crossings), PFD increases.
- If the signal becomes smooth and periodic (fewer zero crossings), PFD decreases.
- Seizure EEG usually shows **reduced PFD** because it becomes repetitive and rhythmic.

Summary

- **HFD** captures complexity based on multi-scale curve length.
- **PFD** captures complexity based on zero-crossing statistics.
- Both decrease when EEG becomes more regular — a key indicator of seizure onset.

Section 8 — Autoregressive (Burg) Features

Autoregressive (AR) modeling expresses each EEG sample as a weighted sum of previous samples. AR features are powerful because seizure activity tends to become more **predictable** and **periodic**, which strongly affects AR coefficients and prediction error.

8.1 AR Model Definition

A p-th order AR model represents the signal as:

$$x[n] = -\sum_{i=1}^{p} a_i \, x[n-i] + e[n]$$

Where:

- a_i = AR coefficients (model parameters)
- $e[n]$ = prediction error (residual)

Interpretation:

- If the signal is rhythmic (like in a seizure), it can be approximated well using only a few past samples.
- Thus the prediction error $e[n]$ becomes **small**.

8.2 Burg Algorithm

The Burg algorithm is used because it:

- minimizes both forward and backward prediction errors
- guarantees a stable AR model
- works well for EEG signals

Burg uses:

- Forward prediction errors $ef[n]$
- Backward prediction errors $eb[n]$

The reflection coefficient at order m is:

$$k_m = \frac{-2\sum_{n=m}^{N-1} ef[n]\, eb[n-1]}{\sum_{n=m}^{N-1}(ef[n]^2 + eb[n-1]^2)}$$

These k_m values update the AR coefficients a_i.

8.3 Outputs from Burg AR

The algorithm outputs:

- **AR coefficients** $a_1, a_2, \ldots, a_p$
- **Prediction error** E

Interpretation:

- Larger $E \rightarrow$ signal is irregular or noisy.
- Smaller $E \rightarrow$ signal is rhythmic/predictable (common in seizures).

Summary

AR features help detect seizures by measuring how well the EEG signal can "predict itself." Seizures tend to produce:

- More rhythmic EEG patterns
- Lower AR prediction error
- Distinct AR coefficient patterns

These features are widely used in biomedical DSP and clinical EEG analysis.

Section 9 — MFCC-Like Cepstral Features

MFCC-like features measure the **overall spectral shape** of the EEG signal. Although MFCCs were originally designed for speech, they work extremely well for EEG because they capture:

- broad spectral changes,
- energy distribution across frequency bands,
- rhythmic patterns typical of seizure onset.

In this system, the MFCC pipeline is adapted to the EEG range (0.5–70 Hz).

9.1 Processing Steps

The MFCC-like extraction follows these stages:

1. **Compute the Power Spectral Density (PSD)** Using the 512-point FFT:
$$P[k] = \Re[k]^2 + \Im[k]^2$$
2. **Warp frequencies to a logarithmic scale (0.5–70 Hz)** EEG rhythms spread multiplicatively rather than linearly:
$$f_{\log} = \log(f)$$
3. **Apply 13 triangular filter banks on the log-frequency axis** If $B = 13$ filters, each filter extracts the average energy in its band.

4. **Compute log energies of each band** For band energy $E[n]$:
$$\text{logE}[n] = \log(E[n] + \epsilon)$$
(small ϵ avoids $\log(0)$)
5. **Apply DCT-II to decorrelate bands (cepstral transform)** This transforms spectral shape into compact coefficients:
$$C[m] = \sum_{n=0}^{N-1} \text{logE}[n] \cos\left(\frac{\pi m(n+\frac{1}{2})}{N}\right)$$
where:
   - $C[m]$ = MFCC-like coefficient,
   - N = number of filter banks (13),
   - m = coefficient index.

9.2 Interpretation

- The first coefficient ($C[0]$) measures overall spectral energy.
- Higher coefficients capture curvature and shape of the PSD.
- Seizures often alter the spectral envelope, causing sudden changes in MFCC values.

MFCC-like features provide a compact, highly discriminative representation of EEG spectral structure.

Section 10 — Seizure Detection Algorithm

The system uses a **hybrid rule-based** seizure detector that combines multiple EEG biomarkers. Each 1-second block of preprocessed data produces a set of features, and the rules evaluate whether the signal exhibits seizure-like properties.

10.1 Components of the Hybrid Detector

The detector monitors several independent indicators:

- **Amplitude-based criteria** Large bursts in EEG cause the Root Mean Square (RMS) and Variance to rise:
$$\text{RMS} > \text{threshold}, \qquad \sigma^2 > \text{threshold}$$
- **Entropy suppression** Seizures often become highly rhythmic and predictable, reducing Shannon entropy:
$$H_{\text{Shannon}} < H_{\min}$$
- **Spectral shifts** The dominant frequency suddenly increases, or the spectral centroid moves upward.
- **MFCC distortions** MFCC-like cepstral coefficients change abruptly when the spectral envelope becomes peaked or narrow.

If any of these conditions are satisfied, the block is considered *seizure-like*.

10.2 Hysteresis for Temporal Stability

To avoid rapid switching between "NORMAL" and "SEIZURE," the algorithm uses a **hysteresis counter**.

$$\text{if seizure-like: } h \leftarrow h + 1$$

$$\text{else: } h \leftarrow h - 1$$

with limits:
$$0 \le h \le H_{\max}$$

A seizure is declared only when:

$$h \ge H_{\text{trigger}}$$

and cleared only when:

$$h = 0$$

This ensures the output state is stable and robust against noise or brief fluctuations.

Section 11 — UART Telemetry

After processing each 1-second data block, the system transmits all extracted features and the detected state ("NORMAL" or "SEIZURE") to the ESP8266. The communication uses **UART over Serial3** (PC10/PC11) with a robust, framed JSON packet.

11.1 JSON Payload

A complete JSON message contains:

- All time-domain features (RMS, Variance, Hjorth parameters, etc.)
- Spectral features (bandpowers, dominant frequency, centroid, flatness)
- Fractal features (HFD, PFD)
- AR prediction error
- MFCC-like coefficients
- Final seizure state

Example structure:

```
{"rms":0.00123, "var":0.00056, "entropy":3.21, "state":"N
```

11.2 CRC16-MODBUS Integrity Check

To ensure that the JSON is received correctly (especially important for a medical-style system), a **CRC16-MODBUS checksum** is computed over the entire JSON string.

The CRC polynomial is:

$$\text{Poly} = 0xA001$$

The CRC output is appended outside the JSON, ensuring the JSON remains parseable.

11.3 Framed UART Format

The final UART packet is wrapped inside angle brackets:

$$\langle \text{JSON},\ \text{crc} : \text{value} \rangle$$

For example:

<{"rms":0.0012, "var":0.0009, "state":"SEIZURE"}, "crc":54321>

This framing ensures that:

- The ESP8266 can detect packet boundaries,
- The CRC can verify packet integrity,
- Corrupted or partial packets can be safely discarded.

Thus the telemetry layer is reliable, structured, and suitable for continuous real-time EEG monitoring.

Section 12 — EEG Physiology Context

EEG rhythms correspond to coordinated neural activity across different brain regions. Understanding these rhythms is essential for interpreting the features computed by the system and for explaining why certain biomarkers change during seizures.

- **Delta (0.5–4 Hz)** Dominant during deep sleep. Certain epileptic syndromes (e.g., Lennox–Gastaut) produce rhythmic delta activity during seizures.
- **Theta (4–8 Hz)** Common in drowsiness, but also appears in many *focal onset seizures*. Sudden, strong theta rhythmicity is a known seizure pattern.
- **Alpha (8–13 Hz)** The primary "resting" rhythm of the occipital cortex. During most seizures, alpha activity **drops sharply** (desynchronization).
- **Beta and Gamma (13–30 Hz, 30–70 Hz)** Fast oscillations increase during ictal bursts. These high-frequency components reflect intense synchronous firing of neural circuits.
- **Shannon Entropy** Measures unpredictability. During seizure onset, the EEG becomes highly rhythmic and predictable, causing entropy to:

$$\text{Entropy} \downarrow$$

- **Fractal Dimension (HFD, PFD)** EEG normally has complex, scale-invariant structure. During seizures, the waveform becomes more periodic and less complex:

$$D_{\text{fractal}} \downarrow$$

- **Autoregressive (AR) Prediction Error** If a signal becomes more periodic, it becomes easier to predict from its past values. Thus, during ictal rhythmic activity:

$$E_{\text{AR}} \downarrow$$

These physiological principles justify why the extracted DSP features reliably indicate the transition from normal to seizure-like EEG activity.

Section 13 — Why Cortex-M4F Works

The STM32F446RE is based on the ARM Cortex-M4F core, which is specifically designed for real-time digital signal processing on low-power embedded systems. Several architectural features make it ideal for EEG processing:

- **Hardware Floating-Point Unit (FPU)** The M4F includes a single-precision FPU that accelerates operations such as:

$$+,\ -,\ \times,\ \div,\ \sqrt{\ }$$

  This is crucial because FIR filters, FFTs, and feature extraction depend heavily on floating-point math. The FPU allows:
  - faster execution,
  - lower CPU load,
  - more accurate signal representation.
- **DSP SIMD Instructions** The M4F has built-in DSP extensions, including:
  - single-cycle Multiply-Accumulate (MAC),
  - SIMD (Single Instruction, Multiple Data) operations,
  - saturating arithmetic.

  These instructions are used in:
  - FIR convolution loops,
  - vector dot products,
  - autocorrelation calculations,
  - FFT butterfly stages.
- **Highly Optimized CMSIS-DSP Library** ARM provides the CMSIS-DSP library, which contains:
  - FIR filters,
  - FFT/IFFT routines,
  - statistical functions,
  - interpolation and transform functions.

  All routines are tuned for the Cortex-M4F pipeline and registers.
  CMSIS-DSP uses:

$$\text{radix-4 and radix-8 FFT butterflies}$$

  and unrolled FIR loops for extremely high performance.

Together, these hardware and software features let the Cortex-M4F handle:

- a 301-tap FIR,
- 8 subband FIRs,
- a 512-point FFT,
- AR modeling,
- fractal calculations,
- MFCC extraction,

all within the 1-second processing window required by the 512 Hz sampling rate.

Section 14 — System Data Flow Summary

The complete EEG-processing system follows a strict, real-time pipeline. Each stage transforms the signal closer to a meaningful clinical interpretation while maintaining the timing requirements of a 512 Hz sampling rate.

1. **DMA fills the ADC buffer** Timer TIM2 triggers ADC1 at 512 Hz. The DMA engine writes samples into a double buffer without CPU involvement. Half and full buffer interrupts indicate when new data blocks are ready.
2. **Convert raw ADC samples to floating-point voltages** Each 12-bit sample is scaled:
$$V = \frac{v}{4095} \times 3.3, \qquad x[n] = V - 1.65$$
This yields centered EEG values in volts.
3. **Apply 50 Hz notch filtering** A biquad IIR filter removes mains interference using:
$$H(z) = \frac{b_0 + b_1 z^{-1} + b_2 z^{-2}}{1 + a_1 z^{-1} + a_2 z^{-2}}$$
4. **Apply 0.5–70 Hz FIR bandpass filtering** A 301-tap FIR (Kaiser window) removes DC drift and high-frequency noise, leaving only the physiologically relevant EEG range.
5. **Perform 8-way subband FIR filtering** Each 129-tap FIR isolates a canonical EEG rhythm (Delta, Theta, Alpha, Beta, Gamma). These subbands provide clearer energy and feature estimates.
6. **Compute all time, frequency, fractal, AR, and cepstral features** Features include:
RMS, σ^2, Hjorth, Bandpowers, Entropy, HFD, PFD, AR error, MFCC-like,
extracted from each 1-second block.
7. **Run the hybrid seizure detector** Multiple biomarker rules evaluate the block:
   - RMS/variance thresholds,
   - entropy suppression,
   - spectral shifts,
   - MFCC distortions.

   Hysteresis ensures stable seizure state decisions.
8. **Send JSON + CRC16 frame over UART** The final packet:
$$\langle \text{JSON},\ \text{crc} : \text{value} \rangle$$
is transmitted through Serial3 (PC10/PC11) for the ESP8266 to display or forward.

This summarizes the complete end-to-end flow from raw EEG acquisition to reliable seizure-state transmission.

Section 15 — ESP8266 EEG Receiver Architecture

The ESP8266 acts as the network-enabled companion to the STM32. It performs user-interface rendering, JSON feature decoding, alert generation, and cloud notifications. Its responsibilities can be divided into:

- Reliable UART reception from STM32
- Real-time dashboard visualization
- Local buzzer/LED alarm output
- Telegram alert messaging
- Log storage + CSV export
- Themed web interface with charts

The ESP8266 does **no DSP**. It accepts already-processed JSON frames from STM32.

Section 16 — UART Communication Layer

16.1 SoftwareSerial

STM32 sends framed messages:

$$\langle \text{JSON} \rangle$$

The ESP listens on:

$$\text{RX} = D2, \quad \text{TX} = D1, \quad \text{baud} = 9600$$

A message begins when the character "¡" arrives and ends when the character "¿" arrives.

16.2 Handshake Protocol

At boot:

1. ESP waits for the string `"HELLO_ESP"`.
2. Once received, ESP responds with `"Response"`.
3. The STM32 then starts transmitting feature JSON packets.

This prevents unsynchronized UART data at startup.

Section 17 — JSON Packet Handling

17.1 Circular Buffer Storage

The ESP maintains a rolling buffer:

$$\text{MAX_MSGS} = 60$$

For each message:

- the raw JSON string is stored,
- a timestamp (millis) is stored,
- a seizure boolean is computed,
- dominant frequency is extracted.

17.2 JSON Parsing (Manual)

Due to memory constraints, ESP8266 avoids using ArduinoJSON. Instead, fields are parsed by string search:

$$\text{value} = \texttt{substring(colon+1, end)}$$

This keeps RAM usage below 20 kB.

Section 18 — Local Alarm System

When a JSON packet contains the word "SEIZURE":

- buzzer oscillates 3 times,
- LED blinks,
- screen flash animation is triggered,
- Telegram alert is sent.

This creates both **visual** and **auditory** cues.

Section 19 — Telegram Cloud Alert Mechanism

The ESP sends alerts via:

```
api.telegram.org/botTOKEN/sendMessage
```

TLS is enabled using:

```
WiFiClientSecure::setInsecure()
```

This bypasses certificate validation (necessary due to ESP8266's limited flash).

19.1 URL Encoding

Message text is encoded:

$$\%20 = \text{space}, \quad \%2F = /$$

A custom URL-encoder builds safe HTTP requests.

Section 20 — Web Dashboard Architecture

The ESP hosts a modern HTML/CSS/JS dashboard.

20.1 Dashboard Components

- Real-time seizure graph (Chart.js)
- Dominant frequency plot
- Spectrogram (custom pixel buffer)
- Synthesized EEG waveform
- Feature grid (24+ features)
- Log viewer
- CSV export tool

20.2 Dark/Light Theme System

Theme toggle modifies root CSS variables:

```
:root[data-theme='dark'] { ...  }
```

Stored in:

```
localStorage["esp_theme"]
```

so the user's theme preference persists.

Section 21 — Spectrogram Engine (ESP-Side)

A custom rasterized spectrogram is rendered on a 2D canvas.

21.1 Mapping Logic

Bands used:

$$\Delta,\ \Theta,\ ,$$

Mapping amplitude → color intensity uses:

$$I = 255 \cdot \frac{\log_{10}(1 + |P|)}{\log_{10}(1 + P_{\max})}$$

Each new update shifts the entire image left by 1 pixel. The rightmost column is filled with new band powers.

21.2 Color Coding

- Delta: green
- Theta: blue
- Alpha: yellow
- Beta: red

Section 22 — Synthesized Waveform

The ESP reconstructs an approximate waveform using:

$$x(t) = \sum A_i \sin(2\pi f_i t)$$

Where:

$$A_\Delta = \sqrt{\Delta} \times 0.01$$
$$A_\Theta = \sqrt{\Theta} \times 0.01$$
$$A = \sqrt{\ } \times 0.01$$
$$A = \sqrt{\ } \times 0.01$$

The choice of:

$$f_\Delta \approx 2, \quad f_\Theta \approx 6, \quad f \approx 10.5, \quad f \approx 20$$

produces a real-time "animated" synthetic EEG wave corresponding to the incoming band powers.

This helps visually interpret EEG activity.

Section 23 — Real-Time Charts

Two Chart.js plots:

- **Seizure graph:** toggles between 0 and 1
- **Dominant frequency:** tracks peak spectral band (Hz)

Each update is non-animated for speed:

```
chart.update("none")
```

Section 24 — Log Viewer and CSV Export

24.1 Log Viewer

All incoming JSON packets are shown as text.

24.2 CSV Export

Dataset is downloaded as:

```
timestamp,rms,variance,delta,theta,alpha,beta,...
```

Uses:

`Blob()` → `URL.createObjectURL()`

so no cloud storage is needed.

Section 25 — REST Endpoints Implemented

- `/` → dashboard
- `/localdata` → JSON log dump
- `/clear_logs`
- `/trigger_alert`
- `/reset_led`
- `/ping`
- `/heap`

All endpoints respond in JSON or plaintext.

Section 26 — Memory-Constrained Design Considerations

- No ArduinoJSON used (saves RAM)
- No SSL certificate bundle (uses `setInsecure()`)
- All HTML served in chunks via `server.sendContent()`
- Spectrogram stored only as a raster buffer
- Logs limited to 60 entries max

This ensures the 80 kB RAM of ESP8266 is not exceeded.

Section 27 — ESP–STM32 System Integration Summary

1. STM32 processes EEG and sends JSON
2. ESP receives framed packets
3. ESP validates, logs, displays
4. ESP triggers alarms + Telegram alerts
5. Dashboard shows real-time analytics

The ESP8266 acts as the "cloud + UI brain", while STM32 is the "DSP brain".

Section 29 — Complete Project Summary

This project implements a full real-time EEG analysis system using an STM32F446RE (Cortex-M4F) for digital signal processing and an ESP8266 for visualization, networking, and alerts. The system mirrors the architecture of professional EEG acquisition and seizure detection devices while remaining lightweight, embedded, and low-power.

29.1 End-to-End Overview

The complete signal flow is:

Electrodes → AFE (Instrumentation Amplifier) → ADC (512 Hz) –

Each subsystem is designed for reliability, noise robustness, physiological validity, and real-time operation.

29.2 Key Strengths of the System

- **Medical-grade sampling pipeline** TIM2 hardware timer + ADC + DMA double buffering ensures precise, gap-free acquisition at 512 Hz.
- **High-fidelity preprocessing** 50 Hz notch + 0.5–70 Hz FIR bandpass + 8 subband FIRs reproduce the same spectral structure used in clinical EEG monitors.
- **Extensive feature set (25+ biomarkers)** Includes:
  - time-domain (RMS, variance, ZC, crest, kurtosis, skewness),
  - spectral (FFT, bandpowers, centroid, flatness),
  - fractal (HFD, PFD),
  - autoregressive (Burg prediction error),
  - cepstral (MFCC-like coefficients).
- **Hybrid seizure detector** Uses amplitude, entropy, spectral shifts, MFCC deformation, and hysteresis to robustly classify EEG blocks as "NORMAL" or "SEIZURE".
- **Reliable STM32 → ESP8266 communication** Framed UART JSON packets with CRC16-MODBUS integrity checking ensure clean delivery of telemetry.
- **Modern ESP8266 web dashboard** Features:
  - spectrogram visualization,

  - synthetic EEG waveform generation,
  - real-time feature grid,
  - seizure detection traces,
  - CSV export of all data,
  - log viewer,
  - dark/light theme.

- **Cloud + local alerts** The ESP8266 triggers:
  - LED/buzzer alerts,
  - Telegram push notifications.

29.3 Performance Characteristics

- Entire DSP pipeline runs comfortably within the 1-second real-time budget.
- CMSIS-DSP optimizations exploit FPU and SIMD for high throughput.
- Memory usage fits within STM32F446RE and ESP8266 constraints.
- Packet reliability ensured via CRC16 and framed UART messages.
- Dashboard performs smooth real-time visualization at 1 Hz updates.

29.4 System as a Whole

The system achieves:

Acquisition+DSP+Feature Extraction+Seizure Detection+Telemetry+Visualization+Alerts

all running continuously and autonomously on microcontroller hardware.

It demonstrates a full-stack biomedical engineering solution with:

- embedded DSP,
- signal theory,
- systems engineering,
- UI/UX design,
- networking and cloud integration.

29.5 Final Takeaway

This project is a complete miniature EEG monitoring platform. It is capable of:

- reliably collecting EEG signals,
- applying clinically meaningful DSP filters,
- computing rich EEG biomarkers,
- identifying seizure-like activity,
- and presenting it through a modern web-based interface.

It combines electrical engineering, embedded systems, signal processing, biomedical engineering, and software/UI development into a coherent, end-to-end system suitable for research, prototyping, or demonstration.

Section 30 — Limitations & Reliability Considerations

Although the system is powerful and fully functional, it is still an embedded, prototype-level EEG processing platform. To understand its boundaries and realistic performance expectations, we must consider several categories of limitations.

30.1 Hardware Limitations

(a) ADC Resolution (12-bit) With a resolution of:

$$2^{12} = 4096 \text{ levels}$$

the smallest step is:

$$\Delta V \approx 0.8 \text{ mV}$$

Clinical EEG machines typically use 16–24 bit ADCs with resolutions of:

$$0.05\text{–}0.5\ \mu\text{V}$$

Thus:

- small microvolt EEG fluctuations may be partially masked,
- front-end instrumentation amplifier must be high gain and low noise,
- precision depends heavily on analog AFE performance.

(b) Single Channel The current design processes only one EEG channel. Clinical systems use:

$$8 \text{ to } 32 \text{ channels}$$

which help in:

- spatial localization of seizures,
- artifact rejection,
- multi-electrode correlation analysis.

(c) Floating-point Throughput Limits The Cortex-M4F FPU is fast, but not medical-grade:

- very long FIR filters,
- multi-channel FFTs,
- machine-learning models

may exceed real-time constraints at 512 Hz unless optimized.

(d) ESP8266 Memory The ESP8266 has:

$$\approx 80 \text{ kB RAM}$$

Therefore:

- No large JSON libraries,
- No certificate bundle storage,
- No machine learning inference,
- Limited dashboard history depth.

30.2 Software Limitations

(a) Manual JSON Parsing String-based JSON extraction is lightweight but fragile:

- malformed JSON may cause failed field parsing,
- nested objects or arrays are not supported,
- improper escaping can corrupt CSV or logs.

(b) Bounded Log Buffer The message buffer stores:

$$\text{MAX_MSGS} = 60$$

meaning:

- only the most recent 1 minute of data is stored,
- long-term trends must be downloaded via CSV,
- no persistent storage (SD card) is available.

(c) No Packet Retransmission UART frames include CRC16 but:

- corrupted frames are dropped,
- no retry mechanism exists,
- high noise on lines may reduce reliability.

(d) Dashboard Rendering Constraints JavaScript runs on the browser:

- some animations disabled (for speed),
- real-time update rate fixed at 1 Hz,
- no offline mode.

30.3 Signal Processing Limitations

(a) Single FIR Pipeline Only one 301-tap FIR is used for wide bandpass. Clinical systems sometimes use:

- adaptive filters,
- ICA/CCA for artifact removal,
- wavelet denoising.

(b) No Artifact Rejection Real EEG contains:

- eye-blinks,
- EMG activity,
- electrode pops,
- motion artifacts.

The current system:

- does not detect artifacts,
- does not suppress them,
- may confuse artifacts with seizures.

(c) Limited Frequency Resolution With:

$$N = 512, \quad f_s = 512\,\text{Hz}$$

the FFT resolution is:

$$\Delta f = 1 \text{ Hz}$$

Clinical devices often use:

$$0.25 \text{ Hz resolution}$$

for finer spectral interpretation.

(d) Rule-Based Seizure Detector While effective, it has limitations:

- no machine learning,
- threshold tuning is manual,
- cannot distinguish seizure types,
- can produce false positives for rhythmic artifacts.

30.4 Network Alerting Limitations

(a) setInsecure() TLS Mode Telegram messages use insecure TLS mode:

```
client.setInsecure()
```

which disables certificate validation (necessary on ESP8266).

Thus:

- vulnerable to MITM if on an untrusted network,
- not suitable for medical deployments.

(b) No Offline Storage of Alerts If WiFi disconnects:

- alerts cannot be queued,
- network interruptions silently skip messages.

(c) Limited Dashboard Responsiveness Updates occur every 1 second, not continuous streaming. This is sufficient for EEG block processing but slower than ICU systems.

30.5 Safety Reliability Considerations

(a) Not a Clinical Device Despite strong engineering, this is not a medical-grade system:

- no FDA/CE compliance,
- no certified electrodes,
- not galvanically isolated from patient,
- no medical-grade ADC.

(b) Electrical Safety Electrode circuits MUST be isolated using:

- medical isolation amplifiers,
- optocouplers,
- battery-only operation,
- proper shielding.

(c) False Positives / False Negatives The hybrid rule detector may:

- trigger false alarms during strong artifacts,
- miss subtle seizure onset patterns,
- produce delayed detection.

(d) No Long-Term Storage or Review Unlike clinical EEG machines, this prototype does not:

- store raw EEG,
- provide long-term trend plots,
- enable clinician review tools.

30.6 Overall Reliability Assessment

- For engineering demonstration, reliability is high.
- For research prototyping, reliability is moderate.
- For clinical deployment, reliability is not adequate without major isolation, filtering, and algorithmic upgrades.

This is a robust engineering prototype, not a medical device.